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Section:	**S18**	Grade:	

1.0 Unix / Linux Command Line Interface (CLI) Basics

This section of the course provides a discussion and hands-on guide on the basic CLI commands in Unix/Linux. The basic CLI commands are: how to log-in, log-out, interpreting the command prompt, and how to enter commands. The objectives of this section are:

- Be able to login remotely to a Unix/Linux server
- Be able to enter commands at the command prompt

1.1. Logging In

A Unix/Linux server can be accessed remotely using the Telnet or SSH protocol. Normally, a terminal emulator software that supports Telnet or SSH (Secure Shell) protocol can be installed on a computer to access a remote server. Telnet is an old unsecure protocol while SSH is a newer and more secure protocol. SSH is used more often these days as opposed to Telnet. PuTTY is a terminal emulator that can downloaded at https://www.putty.org and installed on a MS Windows computer.

Before you try to log-in to a Unix / Linux server, ask your instructor for a user account, the IP address or server name and port number where your account is valid. On your Windows operating system, open the PuTTY terminal emulation software then enter the IP address or name of the Linux server in the “Host Name” text box and port number in the “Port” text box (by default SSH protocol port number is 22 but there are servers that uses a different number), as shown in Fig. 1.1. Press the “Open” button to connect to the server.

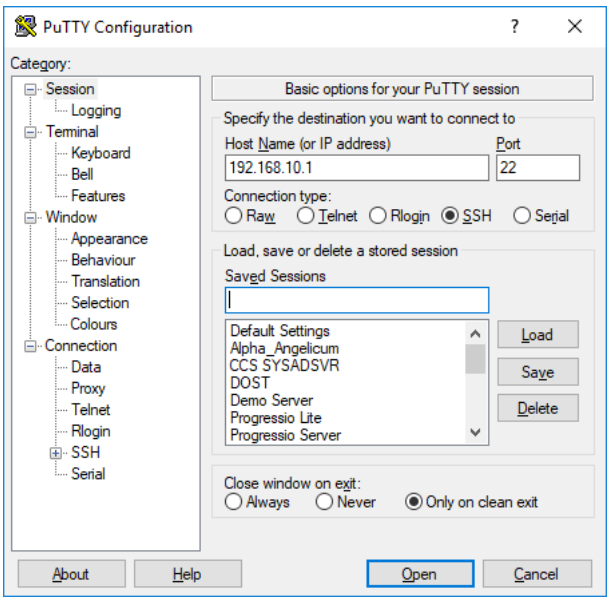


Fig. 1.1 Sample PuTTY Terminal connection

When connected to the server, you will be greeted with login prompt that looks like in Fig. 1.2:

```
Login: <username>
Password:
```

Fig. 1.2. Login prompt in Putty when connected to a server

Step 1 Write down the IP address, port number and user account assigned to you by your instructor.

Server IP address	:	ccscloud.dlsu.edu.ph
Server port number	:	2111
Assigned username/password	:	lbysyads18user20 QsviRNWU

Step 2 Open the PuTTY terminal emulator (or any terminal emulator). Input the IP address and port number of the sever in the appropriate text box. Press the "Open" button to connect to the Linux server. At the login prompt, type your assigned username and password.

2a. Were you able to login? How were you able to login?

Yes, I logged in by typing my assigned username and password. The password may not show on screen for security, but it is still being recorded every time I type a character.

2b. Did you have a user account when you logged-in?

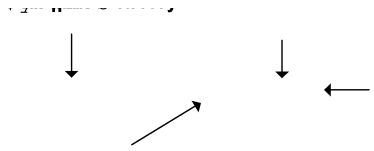
Yes, I used the user account provided by my instructor

2c. What is the IP address or server name of the Unix server you logged into?

I logged into ccscloud.dlsu.edu.ph . This is DLSU's domain for its cloud service.

1.2. The Command Prompt

After you have logged-in, notice that a prompt will appear on the screen. A prompt is an interface that allows user to input commands. Different shells use different faces of prompts (shells will be tackled later in the course). In this server, you will be greeted with:



Notice the different parts of your prompt. For the example prompt, it has displayed the username, current directory you are in, the computer name and the prompt sign.

Step 3 Write down the prompt on your screen.

lbysyads18user20@SYSSVR1:~$

3a. Is your username displayed on the prompt? What is your username?

Yes it is displayed. My username is lbysyads18user20

3b. Is a computer name displayed on the prompt? What is the computer name?

Yes. The computer name is SYSSVR1

3c. What is the prompt symbol on your screen?

The prompt symbol is ~$

1.3. Entering Commands

Commands can be entered to the Unix server via the command prompt. This is usually done by typing a series of characters after the *prompt sign*. A usual format in entering commands is:

```
user01@sysadsvr:~$  command <options> <argument>
```

Options for the command are switches that tell a command to do something differently. Arguments are information sometimes needed by the command to be able to execute properly. Most of the time, options and arguments are not needed in a command.

There are also commands wherein you do not need to type anything. These commands are: ctrl-c, ctrl-d, ctrl-s, and ctrl-d. The table below gives a short description.

Command	Description
ctrl-c	Terminates or aborts a process
ctrl-d	Sends an end-of-input on the keyboard
ctrl-s	Pauses screen output
ctrl-q	Continues screen output

Step 4 On the command prompt, type "`man -S 1 pwd`" and press enter. This command will be tackled later in the course, for now just accept it. ☺

4a. What is the option in the sample `man` command?

-S is the option. Its purpose is to restrict the output to a certain section

4b. What is the argument in the sample `man` command?

1 pwd is the argument. 1 refers to the section number and pwd is the name of the manual page we are trying to view.

4c. Press "q" to exit the command.

Step 5 On the command prompt, type in the command "`ls -a`". This command shows files in your account.

5a. Was the command successful? What was the output?

Yes the command is successful. Below is the output, the files in blue color are directories/folders

Step 6 On the command prompt, type in the command "`LS -a`". This command was same as before but using capital letters.

6a. Was the command successful? Why do you think it was not successful?

The command was not successful, this is due to the case sensitivity of linux commands. LS is not the same as ls.

Step 7 On the command prompt, type "`top`" and press enter.

7a. Press the control and "c" key at the same time. What happened? Did it have the same effect as pressing "q" (You may retry the "top" command and press "q" this time to check)? Why?

Yes they both have the same effect. Both commands bring out the prompt, lbysyads18user20@SYSSVR1:~$

Ctrl-c is for aborting a process while Ctrl-q is for continuing screen output. They have the same output because they are quite similar commands. Though, the key difference is 'ctrl-c' sends an interrupt signal to a process while 'q' is more of an internal quit function in a specific program. Both of which lead to the same prompt.

Step 8 On the command prompt, type "`man chmod`" and press enter.

8a. Press the spacebar. Did the screen scroll up?

Yes the screen scrolled up

8b. Press the control and "f" key at the same time and then press the spacebar. Did the screen scroll up? Why?

No. The screen didn't move because there is no such command for ctrl-f-spacebar

8c. This time press the control and "b" key at the same time. What happened?

The screen scrolled down back to the previous part.

8d. What does the command "ctrl-f" and "ctrl-b" do?

Ctrl-f scrolls the screen up, similar to spacebar. While Ctrl-b scrolls the screen down

1.4. Changing Password

One of the first things you must do on your user account is to change your password. Password ensures that no one will have unauthorized access to your user account. A user must have to be authenticated using a password before access to server will be given. To change your password, use the command: `passwd`.

You will be prompted by the server of your old password and then will be prompted to input a new password. You will have to re-type your new password for verification. You can change your password anytime. Password must have at least six characters with any alphanumeric combination (at least for Linux) but is case sensitive.

Step 9 At the command prompt, type "passwd".

9a. Enter your current password when prompted for the old password.

9b. Enter the password "`h3llo`", when prompted for the new password.

9c. Was the password rejected? Why was it rejected?

Yes the password was rejected because it was too short. It should have at least 6 characters.

9d. Enter the password "`r3ch3tt`" when prompted for the new password.

9e. Was the password was accepted? Why was it accepted?

Yes the password was accepted. This is because it meets the 6 character minimum requirement.

9f. What is the message when the password was successfully changed?

The message is "passwd: password updated successfully"

Step 10 At the command prompt, type "`passwd`" again. Type the current password as "`R4ch3tt`".

10a. Was the current password rejected? Why was it rejected?

The password was rejected because it does not match with the password we set earlier.

10b. What is the message when the password was not successfully changed?

The message is
passwd: Authentication token manipulation error
passwd: password unchanged

Step 11 This time, use "passwd" to change your desired password. Remember your new password.

11a. Was your new chosen password accepted on your first try? What were the messages from the server?

No, I was prompted to type it a second time. This is for security to ensure the user confirms their new password. The following are the messages from the server
Current password:
New password:
Retype new password:
passwd: password updated successfully

1.5. Logging Out

The "exit" command can be used to log out of the Unix server.

At the prompt type the command "exit".

11b. What happened? Were you able to log out? Is there an alternative command for this? (You may research this on books or on the Internet)

Yes, I was able to log out. Ctrl-d is another command to exit the session, it exits by sending an end of file character.

1.6. Conclusion/Reflection

What are some of the important things you learned while typing and entering commands on the terminal in this activity?

The most important thing I learned is that case sensitivity is a vital thing to note when entering commands. It is different compared to no-case-sensitivity nature of Cisco switches and routers. To ensure we get the desired output, we should make sure we type it in the right case. I also realized that some linux commands have many alternatives that lead to the same thing, I should note these alternatives so that I am familiar and would not get confused.

What is the difference between remote access vs. accessing the server machine physically? What are the advantages and disadvantages of using remote access?

Remote access allows us to control a server from a far location through a device, whereas physical access to a server requires actual physical presence and direct interaction with the hardware.

Remote access is beneficial because it is convenient in limiting travel costs. This also offers flexibility and the administrator can control the server in the comfort of his/her own home. It is also helpful for companies as they can employ people globally to handle their server, offering a wider pool of talent.

Though, remote access can be challenging because it relies on a network connection. When this connection is lost, access to the server is totally gone. The speed of the connection may also cause lags, glitching, and inefficient performance. Security is also at risk especially if the connection is not secure, threats to malware, hacking, or phishing is possible.